Bohan Zhou

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EDUCATION

The Chinese University of Hong Kong, Computer Science and Engineering — Hong Kong, China
PhD in Computer Science and Engineering — *Aug. 2026 – Present*

- *Tsinghua 3rd Embodied AI Challenge* — Gripper Track, **3rd Prize**.

Peking University, School of Computer Science — Beijing, China
Master in Computer Applied Technology (GPA: 3.88 / 4.0) — *Sep. 2023 – Jul. 2026*

Nankai University — Tianjin, China
Bachelor in Intelligent Science and Technology (GPA: 3.95 / 4.0, Rank: 2/98) — *Sep. 2019 – Jul. 2023*

PUBLICATIONS

[1] **Bohan Zhou**, Ke Li, Jiechuan Jiang, et al. Learning from visual observation via offline pretrained state-to-go transformer. *NeurIPS 2023*.

[2] **Bohan Zhou**, Yi Zhan, Zhongbin Zhang, et al. MEgoHand: Multimodal Egocentric Hand-Object Interaction Motion Generation. *NeurIPS 2025*.

[3] **Bohan Zhou**, Zhongbin Zhang, Jiangxing Wang, et al. NOLO: Navigate Only Look Once. *IROS 2025 Oral*.

[4] **Bohan Zhou**, Haoqi Yuan, Yuhui Fu, et al. Learning diverse bimanual dexterous manipulation skills from human demonstrations. *AAAI 2026 Oral*.

[5] Haoqi Yuan, **Bohan Zhou**, Yuhui Fu, et al. Cross-embodiment dexterous grasping with reinforcement learning. *ICLR 2025*.

[6] Sipeng Zheng, **Bohan Zhou**, Yicheng Feng, et al. Unicode: Learning a unified codebook for multimodal large language models. *ECCV 2024*.

[7] Hao Luo, **Bohan Zhou**, Zongqing Lu. Pre-trained Visual Dynamics Representations for Efficient Policy Learning. *ECCV 2024*.

[8] Weihao Tan, Wentao Zhang, Xinrun Xu, Haochong Xia, Ziluo Ding, Boyu Li, **Bohan Zhou**, et al. Cradle: Empowering foundation agents towards general computer control. *ICML 2025*.

[9] Zhengxi Hu, Yuxue Yang, Xiaolin Zhai, Dingye Yang, **Bohan Zhou**, et al. GFIE: A Dataset and Baseline for Gaze-Following from 2D to 3D in Indoor Environments. *CVPR 2023*.

[10] Zhengxi Hu, ..., **Bohan Zhou**, et al. Gaze target estimation inspired by interactive attention. *TCSVT 2022*.

[11] **Bohan Zhou**, et al. GroundingPI: A Grounding Foundation Model towards Physical Intelligence with Visual Primitives. *(Hugging Face Daily Papers #1; Under Review)*.

[12] Qize Yu, ..., **Bohan Zhou**, ..., et al. AffordanceVLA: A Vision-Language-Action Model Empowering Action Generation through Affordance-Aware Understanding. *ECCV 2026 Workshop*.

[13] Haoqi Yuan, Yu Bai, Yuhui Fu, **Bohan Zhou**, et al. Being-0: A Humanoid Robotic Agent with Vision-Language Models and Modular Skills. *(Under Review)*.

[14] **Bohan Zhou**, et al. FETCH: Frame-grounded Evidence Topology for Embodied Spatial Planning with The Counterfactual Hypothesis. *(Project Lead, Under Review)*.

[15] **Bohan Zhou**, et al. RoboAware: Learning to Coordinate Embodied Skills from Counterfactual Outcomes. *(Project Lead, Under Review)*.

INTERNSHIP EXPERIENCE

Being-0: A Humanoid Robotic Agent with Vision-Language Models and Modular Skills
Beijing Academy of Artificial Intelligence, Embodied AI *Oct. 2024 – Feb. 2025*

- **A hierarchical robot agent framework** for efficient long-horizon humanoid robot control: **Top-level** black-box large multi-modal model (LMM, like GPT-4o) for task understanding + decomposition; **Mid-level** finetuned LMM for navigation planning; **Low-level** skill libraries for dexterous manipulation.
- **A VLM + rule based hybrid connector**: bridged high-level language plans with low-level motor skills, enabling coordinated locomotion, navigation and dexterous manipulation.
- **Real-world deployment**: Deployed on **Unitree H1**; completed **tens-of-meters** long-horizon navigation culminating in a **coffee-delivery** task.

Cradle: Empowering Foundation Agents towards General Computer Control
Beijing Academy of Artificial Intelligence, Generalist Agents *Oct. 2023 – Feb. 2024*

- **General computer control**: Pioneered a universal interface for agents to interact with any software using screenshots as input and keyboard/mouse actions as output, standardizing cross-environment interaction.
- **A LMM-powered Cradle framework**: integrated six core modules (Information Gathering, Self-Reflection, Task Inference, Skill Curation, Action Planning, and Memory) to automate task planning and execution via generating keyboard/mouse codes without built-in APIs.
- **Robust validation**: adaptable across **4 games** (Red Dead Redemption 2, Cities: Skylines, Stardew Valley and Dealer's Life 2), **5 software tools** (Chrome, Outlook, Feishu, Meitu and CapCut), and **OSWorld**.

RESEARCH EXPERIENCE

GroundingPI: A Grounding Foundation Model towards Physical Intelligence with Visual Primitives
Hugging Face Daily Papers #1; Under Review *2026*

- **Grounding Foundation Model**: A **4B** model represents points and boxes as quantized coordinates in a shared vocabulary for precise, language-conditioned spatial perception.
- **Perceptual Evaluation**: Achieved a **73.68%** average across **34 grounding benchmarks** spanning **11 perceptual capabilities**, outperforming **44 baselines**, including GPT-6 Astra (71.54%).
- **VLA Transfer**: As a visual backbone, led all four evaluated manipulation OOD settings.

MEgoHand: Multimodal Egocentric Hand-Object Interaction Motion Generation
BeingBeyond, Embodied AI *Mar. 2025 – May. 2025*

- **Large-Scale Dataset**: Curated **3.35M** hand-object interaction samples, **1.2K** objects, **24K** trajectories (10-15x prior datasets) via proposed inverse MANO retargeting and virtual RGB-D rendering.
- **Multimodal Perception & Spatial Understanding**: Based on **Eagle-2**, MEgoHand additionally incorporates **Unidepth-v2** for metric depth estimation.
- **Smooth Motion Generation**: **DiT**-based motion generator supervises future hand motion via Flow Matching. **Temporal orthogonal filtering** is designed to decode smooth pose sequence.

BiDexHD: Learning Diverse Bimanual Dexterous Manipulation Skills from Human Demonstrations
Beijing Academy of Artificial Intelligence, Embodied AI *Jun. 2024 – Oct. 2024*

- **BiDexHD Framework**: Developed a unified framework for learning diverse bimanual skills from human demonstrations, unifying **task construction from HOI datasets** and **teacher-student policy learning** for scalable vision-based bimanual dexterous skills. To avoid task-specific reward engineering, a two-stage reward function is generally designed.
- **Zero-shot adaptability**: Achieved **74.59%** task fulfillment on trained tasks and **51.07%** on unseen tasks in the TACO benchmark (141 tasks), with **80.49%/65.99%** on ARCTIC.

NOLO: Navigate Only Look Once
Beijing Academy of Artificial Intelligence, Embodied AI *Feb. 2024 – May. 2024*

- **Video Navigation**: Introduced a novel task requiring agents to finish image navigation using only a single **30-second** context video and real-time egocentric images.

- **NOLO Method**: Enhanced offline reinforcement learning by integrating **GMflow** via pseudo-action labeling and a **temporal coherence** loss.
- **Simulation and Real World Evaluation**: Demonstrated success in RoboTHOR and Habitat simulation and validated **real-world deployment on a Unitree Go2** robot in a maze environment.

STG: Learning from Visual Observation via Offline Pretrained State-to-Go Transformer
Beijing Academy of Artificial Intelligence, Generalist Agents *Sep. 2022 – May. 2023*

- **Two-stage framework to learn from pixels**: upstream offline pretrained State-to-Go Transformer on visual observations to guide **reward-free** online reinforcement learning downstream.
- **Joint Representation Learning**: Co-optimized a discriminator and temporal distance regressor in an **adversarial** manner to align latent embeddings temporally.

PVDR: Pre-trained Visual Dynamics Representations for Efficient Policy Learning
Beijing Academy of Artificial Intelligence, Embodied AI *2024*

- **PVDR [ECCV 2024]**: Proposed pretraining visual transfer representations from large-scale unlabeled Internet videos to overcome domain gaps and adapt to downstream reinforcement learning tasks.

UniCode: Learning a Unified Codebook for Multimodal Large Language Models
Beijing Academy of Artificial Intelligence, Generalist Agents *2024*

- **UniCode [ECCV 2024]**: Proposed an efficient multimodal training paradigm that learns a unified codebook, relaxing the constraints of pure-text codebooks on image generation.

Human Intent Analysis in Indoor Environments for Service Robots Advisor: Prof. Jingtai Liu
Tianjin Key Laboratory of Intelligent Robotics *Mar. 2021 – Aug. 2022*

- **Human Intent Analysis Pipeline**: Involved semi-automatic dataset construction and gaze direction/target estimation for human intention understanding and forcasting, achieving **3rd prize** .
- **GFIE [CVPR 2023]**: Created multi-sensor gaze data collection system (Kinect + laser rangefinder) with novel algorithm for unbiased 2D/3D gaze target annotation via laser spot localization.
- **VSG-IA [TCSVT 2022]**: Proposed graph attention network for automatic gaze behavior detection and human-scene interaction analysis.

Awards & Honors

National Scholarship, Triple-A Outstanding Student Award
Peking University, China *Sep. 2025*

Outstanding Graduates in Nankai University
Nankai University, China *Jun. 2023*

Tianjin Municipal People's Government Scholarship (Top 2%)
Tianjin, China *Nov. 2021*

2nd Prize, National College Student Mathematical Modeling Competition
Chinese Society for Industrial and Applied Mathematics *Oct. 2021*

Honorable Mention, American College Mathematical Contest in Modeling
COMAP *May. 2021*

Gongneng Scholarship of Nankai University (Top 5%)
Nankai University, China *Dec. 2020, Dec. 2022*